

Lab 03 - NAND Only Logic

In this lab, you've learned how to convert arbitrary logical equations into NAND only circuits, and why that might be a good thing.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

We learned how to translate an IC schematic into a usable circuit. We used DeMorgan's law to turn a multi-gate circuit into just a NAND gate circuit. We then used all that information to physically wire the circuit together.

Lab Questions

1 - Write down DeMorgan's Law and the truth tables proving it out.

- $\overline{(A \cdot B)} = \bar{A} + \bar{B}$
- $\overline{(A + B)} = \bar{A} \cdot \bar{B}$

Truth tables:

A	B	$\overline{(A \cdot B)}$	$\bar{A} + \bar{B}$
0	0	1	1
0	1	1	1
1	0	1	1

1	1	0	0
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A	B	$\overline{(A + B)}$	$\overline{A} \cdot \overline{B}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

2 - What is the value in converting circuits to NAND only?

NAND gates are more electrically efficient than any other gate, and so having exclusively NAND gates greatly improves your material efficiency.

3 - How does what you did in lab with the breadboard relate to the FPGA?

We physically set up using an IC and wired a circuit which could be done with only a few lines of code with an FPGA. Physically wiring the gates together with the IC is equivalent to programming those same gates in an FPGA.

Code Submission

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